

Signal processing primitives

Overview

Beyond its causal-inference and survey-statistics surfaces, MORIE exposes a small collection of signal-processing primitives. These are useful for forensic-audio work, biomedical signals, and any analysis that needs spectral or time-frequency methods adjacent to the causal pipeline (e.g. inter-incident time series in criminological data).

The R surface is intentionally thin — the heavy lifting lives in the Python `morie.signal_processing` and `morie.homomorphic_deconvolution` modules. The R wrappers cover the most commonly needed primitives.

A synthetic two-tone signal

```
library(morie)

set.seed(42)
fs <- 1000                                # sampling rate (Hz)
t <- seq(0, 1, by = 1 / fs)
sig <- sin(2 * pi * 50 * t) +
      0.5 * sin(2 * pi * 120 * t) +
      0.3 * rnorm(length(t))
```

Basic FFT

```
spec <- stats::fft(sig)
n <- length(sig)
freq <- (0:(n / 2 - 1)) * fs / n
mag <- Mod(spec)[1:(n / 2)]
peak_freqs <- freq[order(mag, decreasing = TRUE)[1:5]]
peak_freqs
#> [1] 49.95005 119.88012 120.87912 50.94905 48.95105
```

The two largest peaks should sit near 50 Hz and 120 Hz, recovering the synthetic signal's components.

Where to go next

- Spectral analysis with windowing, multitaper estimation, and short-time Fourier transforms is implemented in the Python `morie.signal_processing` module — with the same `RichResult` return convention — and is documented in the package paper.

- Homomorphic deconvolution and cepstral methods (used in forensic audio and biomedical-signal applications) live in `morie.homomorphic_deconvolution`.